Topic: Variation of parameters for higher order equations

Question: Use variation of parameters to find the general solution to the differential equation.

$$y''' + y' = \sec t$$

Answer choices:

A $y(t) = c_1 + c_2 \cos t + c_3 \sin t + \ln|\sec t + \tan t| - t \cos t + (\sin t)\ln|\cos t|$

B $y(t) = c_1 e^{-t} + c_2 te^{-t} + c_3 \cos t + \sec^2 t + \csc t - t \sin t$

C $y(t) = c_1 e^{-4t} + c_2 \cos t + c_3 \sin t + \cos t + \sin t + 1$

D $y(t) = c_1 + c_2 \cos t + c_3 \sin t + \ln|\sec^2 t| + t^2 \csc t + 3t^3$

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Solution: A

Our first step is to get the characteristic polynomial and calculate its roots so we can find the complementary solution.

$$r^3 + r = 0$$

$$r(r^2 + 1) = 0$$

The roots are $r_1 = 0$, $r_2 = i$, and $r_3 = -i$, so the complementary solution is

$$y_c(t) = c_1 + c_2 \cos t + c_3 \sin t$$

We now have the fundamental set of solutions $\{1, \cos t, \sin t\}$, and from here we can build the Wronskian determinants.

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 0 & -\cos t & -\sin t \end{vmatrix}$$

$$= 1 \begin{vmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{vmatrix} - 0 \begin{vmatrix} \cos t & \sin t \\ -\cos t & -\sin t \end{vmatrix} + 0 \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix}$$

$$= \sin^2 t + \cos^2 t$$

$$= 1$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \\ 1 & -\cos t & -\sin t \end{vmatrix}$$

$$= 0 \begin{vmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{vmatrix} - 0 \begin{vmatrix} \cos t & \sin t \\ -\cos t & -\sin t \end{vmatrix} + 1 \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix}$$

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$$= \cos^2 t + \sin^2 t$$

$$= 1$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & 0 & \sin t \\ 0 & 0 & \cos t \\ 0 & 1 & -\sin t \end{vmatrix}$$

$$= 1 \begin{vmatrix} 0 & \cos t \\ 1 & -\sin t \end{vmatrix} - 0 \begin{vmatrix} 0 & \sin t \\ 1 & -\sin t \end{vmatrix} + 0 \begin{vmatrix} 0 & \sin t \\ 0 & \cos t \end{vmatrix}$$

$$= 0 - \cos t$$

$$= -\cos t$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} 1 & \cos t & 0 \\ 0 & -\sin t & 0 \\ 0 & -\cos t & 1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} -\sin t & 0 \\ -\cos t & 1 \end{vmatrix} - 0 \begin{vmatrix} \cos t & 0 \\ -\cos t & 1 \end{vmatrix} + 0 \begin{vmatrix} \cos t & 0 \\ -\sin t & 0 \end{vmatrix}$$

$$= -\sin t - 0$$

$$= -\sin t$$

Then the unknowns are

$$u_1' = \frac{g(t)W_1}{W} = \frac{\sec t}{1} = \sec t$$

$$u_2' = \frac{g(t)W_2}{W} = \frac{(\sec t)(-\cos t)}{1} = -1$$

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$$u_3' = \frac{g(t)W_3}{W} = \frac{(\sec t)(-\sin t)}{1} = -\tan t$$

Next, we integrate each of these to get u_1, u_2, and u_3.

$$u_1 = \int \frac{g(t)W_1}{W}\,dt = \int \sec t\,dt = \ln|\sec t + \tan t|$$

$$u_2 = \int \frac{g(t)W_2}{W}\,dt = \int -\,dt = -t$$

$$u_3 = \int \frac{g(t)W_3}{W}\,dt = \int -\tan t\,dt = \ln|\cos t|$$

Then the particular solution is

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \ln|\sec t + \tan t| - t\cos t + (\sin t)\ln|\cos t|$$

The general solution is the sum of the complementary solution and the particular solution,

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1 + c_2\cos t + c_3\sin t + \ln|\sec t + \tan t| - t\cos t + (\sin t)\ln|\cos t|$$

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Topic: Variation of parameters for higher order equations

Question: Use variation of parameters to find the general solution to the differential equation.

$$y''' - 3y'' - 4y' + 12y = t$$

Answer choices:

A $y(t) = c_1e^{-2t} + c_2e^{2t} + c_3e^{3t} + \frac{1}{10}t + \frac{1}{30}$

B $y(t) = c_1e^{-2t} + c_2e^{2t} + c_3e^{3t} + \frac{1}{12}t + \frac{1}{36}$

C $y(t) = c_1e^{3t} + c_2e^{4t} + c_3e^{5t} + 2t + \frac{1}{18}$

D $y(t) = c_1e^{3t} + c_2e^{4t} + c_3e^{5t} + \frac{1}{10}t + \frac{1}{30}$

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Solution: B

Our first step is to get the characteristic polynomial and calculate its roots so we can find the complementary solution.

$$r^3 - 3r^2 - 4r + 12 = 0$$

$$(r + 2)(r - 2)(r - 3) = 0$$

The roots are $r_1 = -2$, $r_2 = 2$, and $r_3 = 3$, so the complementary solution is

$$y_c(t) = c_1 e^{-2t} + c_2 e^{2t} + c_3 e^{3t}$$

We now have the fundamental set of solutions $\{e^{-2t}, e^{2t}, e^{3t}\}$, and from here we can build the Wronskian determinants.

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-2t} & e^{2t} & e^{3t} \\ -2e^{-2t} & 2e^{2t} & 3e^{3t} \\ 4e^{-2t} & 4e^{2t} & 9e^{3t} \end{vmatrix}$$

$$= e^{-2t} \begin{vmatrix} 2e^{2t} & 3e^{3t} \\ 4e^{2t} & 9e^{3t} \end{vmatrix} - e^{2t} \begin{vmatrix} -2e^{-2t} & 3e^{3t} \\ 4e^{-2t} & 9e^{3t} \end{vmatrix} + e^{3t} \begin{vmatrix} -2e^{-2t} & 2e^{2t} \\ 4e^{-2t} & 4e^{2t} \end{vmatrix}$$

$$= e^{-2t}(18e^{5t} - 12e^{5t}) - e^{2t}(-18e^{t} - 12e^{t}) + e^{3t}(-8 - 8)$$

$$= e^{-2t}(6e^{5t}) - e^{2t}(-30e^{t}) + e^{3t}(-16)$$

$$= 6e^{3t} + 30e^{3t} - 16e^{3t}$$

$$= 20e^{3t}$$

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$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & e^{2t} & e^{3t} \\ 0 & 2e^{2t} & 3e^{3t} \\ 1 & 4e^{2t} & 9e^{3t} \end{vmatrix}$$

$$= 0 \begin{vmatrix} 2e^{2t} & 3e^{3t} \\ 4e^{2t} & 9e^{3t} \end{vmatrix} - 0 \begin{vmatrix} e^{2t} & e^{3t} \\ 4e^{2t} & 9e^{3t} \end{vmatrix} + 1 \begin{vmatrix} e^{2t} & e^{3t} \\ 2e^{2t} & 3e^{3t} \end{vmatrix}$$

$$= 3e^{5t} - 2e^{5t}$$

$$= e^{5t}$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-2t} & 0 & e^{3t} \\ -2e^{-2t} & 0 & 3e^{3t} \\ 4e^{-2t} & 1 & 9e^{3t} \end{vmatrix}$$

$$= -0 \begin{vmatrix} -2e^{-2t} & 3e^{3t} \\ 4e^{-2t} & 9e^{3t} \end{vmatrix} + 0 \begin{vmatrix} e^{-2t} & e^{3t} \\ 4e^{-2t} & 9e^{3t} \end{vmatrix} - 1 \begin{vmatrix} e^{-2t} & e^{3t} \\ -2e^{-2t} & 3e^{3t} \end{vmatrix}$$

$$= -(3e^{t} + 2e^{t})$$

$$= -5e^{t}$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^{-2t} & e^{2t} & 0 \\ -2e^{-2t} & 2e^{2t} & 0 \\ 4e^{-2t} & 4e^{2t} & 1 \end{vmatrix}$$

$$= 0 \begin{vmatrix} -2e^{-2t} & 2e^{2t} \\ 4e^{-2t} & 4e^{2t} \end{vmatrix} - 0 \begin{vmatrix} e^{-2t} & e^{2t} \\ 4e^{-2t} & 4e^{2t} \end{vmatrix} + 1 \begin{vmatrix} e^{-2t} & e^{2t} \\ -2e^{-2t} & 2e^{2t} \end{vmatrix}$$

$$= 2 + 2$$

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$$= 4$$

Then the unknowns are

$$u_1' = \frac{g(t)W_1}{W} = \frac{te^{5t}}{20e^{3t}} = \frac{1}{20}te^{2t}$$

$$u_2' = \frac{g(t)W_2}{W} = \frac{-5te^{t}}{20e^{3t}} = -\frac{1}{4}te^{-2t}$$

$$u_3' = \frac{g(t)W_3}{W} = \frac{4t}{20e^{3t}} = \frac{1}{5}te^{-3t}$$

Next, we integrate each of these to get u_1, u_2, and u_3.

$$u_1 = \int \frac{g(t)W_1}{W}\,dt = \int \frac{1}{20}te^{2t}\,dt = \frac{1}{40}te^{2t} - \frac{1}{80}e^{2t}$$

$$u_2 = \int \frac{g(t)W_2}{W}\,dt = \int -\frac{1}{4}te^{-2t}\,dt = \frac{1}{8}te^{-2t} + \frac{1}{16}e^{-2t}$$

$$u_3 = \int \frac{g(t)W_3}{W}\,dt = \int \frac{1}{5}te^{-3t}\,dt = -\frac{1}{15}te^{-3t} - \frac{1}{45}e^{-3t}$$

Then the particular solution is

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(\frac{1}{40}te^{2t} - \frac{1}{80}e^{2t}\right)e^{-2t} + \left(\frac{1}{8}te^{-2t} + \frac{1}{16}e^{-2t}\right)e^{2t} + \left(-\frac{1}{15}te^{-3t} - \frac{1}{45}e^{-3t}\right)e^{3t}$$

$$y_p(t) = \frac{1}{40}t - \frac{1}{80} + \frac{1}{8}t + \frac{1}{16} - \frac{1}{15}t - \frac{1}{45}$$

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$$y_p(t) = \frac{1}{12}t + \frac{1}{36}$$

The general solution is the sum of the complementary solution and the particular solution,

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{-2t} + c_2e^{2t} + c_3e^{3t} + \frac{1}{12}t + \frac{1}{36}$$

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Topic: Variation of parameters for higher order equations

Question: Use variation of parameters to find the general solution to the differential equation.

$$y''' + y'' - y' + 15y = \frac{1}{2}e^{3t}$$

Answer choices:

A $y(t) = c_1e^{4t} + c_2e^{-4t}\cos(4t) + c_3e^{-4t}\sin(4t) + \frac{1}{72}e^{-3t}$

B $y(t) = c_1e^{-2t} + c_2e^{t}\cos t + c_3e^{t}\sin t + \frac{1}{128}e^{2t}\sin(2t)$

C $y(t) = c_1e^{-3t} + c_2e^{t}\cos(2t) + c_3e^{t}\sin(2t) + \frac{1}{96}e^{3t}$

D $y(t) = c_1 + c_2e^{2t}\cos t + c_3e^{2t}\sin t + \frac{1}{56}e^{-2t}$

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Solution: C

Our first step is to get the characteristic polynomial, factor it, and calculate its roots so we can find the complementary solution.

$$r^3 + r^2 - r + 15 = 0$$

$$(r + 3)(r^2 - 2r + 5) = 0$$

The roots are $r_1 = -3$, $r_2 = 1 + 2i$, and $r_3 = 1 - 2i$, so the complementary solution is

$$y_c(t) = c_1 e^{-3t} + c_2 e^t \cos(2t) + c_3 e^t \sin(2t)$$

We now have the fundamental set of solutions $\{e^{-3t}, e^t \cos(2t), e^t \sin(2t)\}$, and from here we can build the Wronskian determinants.

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-3t} & e^t \cos(2t) & e^t \sin(2t) \\ -3e^{-3t} & e^t \cos(2t) - 2e^t \sin(2t) & e^t \sin(2t) + 2e^t \cos(2t) \\ 9e^{-3t} & -3e^t \cos(2t) - 4e^t \sin(2t) & -3e^t \sin(2t) + 4e^t \cos(2t) \end{vmatrix}$$

$$= e^{-3t} \begin{vmatrix} e^t \cos 2t - 2e^t \sin(2t) & e^t \sin(2t) + 2e^t \cos(2t) \\ -3e^t \cos(2t) - 4e^t \sin(2t) & -3e^t \sin(2t) + 4e^t \cos(2t) \end{vmatrix}$$

$$+3e^{-3t} \begin{vmatrix} e^t \cos(2t) & e^t \sin(2t) \\ -3e^t \cos(2t) - 4e^t \sin(2t) & -3e^t \sin(2t) + 4e^t \cos(2t) \end{vmatrix}$$

$$+9e^{-3t} \begin{vmatrix} e^t \cos(2t) & e^t \sin(2t) \\ e^t \cos(2t) - 2e^t \sin(2t) & e^t \sin(2t) + 2e^t \cos(2t) \end{vmatrix}$$

$$= e^{-3t}(10e^{2t}) + 3e^{-3t}(4e^{2t}) + 9e^{-3t}(2e^{2t})$$

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$$= 10e^{-t} + 12e^{-t} + 18e^{-t}$$

$$= 40e^{-t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & e^t\cos(2t) & e^t\sin(2t) \\ 0 & e^t\cos(2t) - 2e^t\sin(2t) & e^t\sin(2t) + 2e^t\cos(2t) \\ 1 & -3e^t\cos(2t) - 4e^t\sin(2t) & -3e^t\sin(2t) + 4e^t\cos(2t) \end{vmatrix}$$

$$= 1\begin{vmatrix} e^t\cos(2t) & e^t\sin(2t) \\ e^t\cos(2t) - 2e^t\sin(2t) & e^t\sin(2t) + 2e^t\cos(2t) \end{vmatrix}$$

$$= 1(2e^{2t})$$

$$= 2e^{2t}$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-3t} & 0 & e^t\sin(2t) \\ -3e^{-3t} & 0 & e^t\sin(2t) + 2e^t\cos(2t) \\ 9e^{-3t} & 1 & -3e^t\sin(2t) + 4e^t\cos(2t) \end{vmatrix}$$

$$= -1\begin{vmatrix} e^{-3t} & e^t\sin(2t) \\ -3e^{-3t} & e^t\sin(2t) + 2e^t\cos(2t) \end{vmatrix}$$

$$= -4e^{-2t}\sin(2t) - 2e^{-2t}\cos(2t)$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^{-3t} & e^t\cos(2t) & 0 \\ -3e^{-3t} & e^t\cos(2t) - 2e^t\sin(2t) & 0 \\ 9e^{-3t} & -3e^t\cos(2t) - 4e^t\sin(2t) & 1 \end{vmatrix}$$

$$= 1\begin{vmatrix} e^{-3t} & e^t\cos(2t) \\ -3e^{-3t} & e^t\cos(2t) - 2e^t\sin(2t) \end{vmatrix}$$

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$$= 4e^{-2t}\cos(2t) - 2e^{-2t}\sin(2t)$$

Then the unknowns are

$$u_1' = \frac{g(t)W_1}{W} = \frac{\left(\frac{1}{2}e^{3t}\right)(2e^{2t})}{40e^{-t}} = \frac{1}{40}e^{6t}$$

$$u_2' = \frac{g(t)W_2}{W} = \frac{\left(\frac{1}{2}e^{3t}\right)(-4e^{-2t}\sin(2t) - 2e^{-2t}\cos(2t))}{40e^{-t}}$$

$$= -\frac{1}{20}e^{2t}\sin(2t) - \frac{1}{40}e^{2t}\cos(2t)$$

$$u_3' = \frac{g(t)W_3}{W} = \frac{\left(\frac{1}{2}e^{3t}\right)(4e^{-2t}\cos(2t) - 2e^{-2t}\sin(2t))}{40e^{-t}}$$

$$= \frac{1}{20}e^{2t}\cos(2t) - \frac{1}{40}e^{2t}\sin(2t)$$

Next, we integrate each of these to get u_1, u_2, and u_3.

$$u_1 = \int \frac{g(t)W_1}{W}\,dt = \int \frac{1}{40}e^{6t} = \frac{1}{240}e^{6t}$$

$$u_2 = \int \frac{g(t)W_2}{W}\,dt = \int -\frac{1}{20}e^{2t}\sin(2t) - \frac{1}{40}e^{2t}\cos(2t)\,dt$$

$$= -\frac{3}{160}e^{2t}\sin(2t) + \frac{1}{160}e^{2t}\cos(2t)$$

$$u_3 = \int \frac{g(t)W_3}{W}\,dt = \int \frac{1}{20}e^{2t}\cos(2t) - \frac{1}{40}e^{2t}\sin(2t)\,dt$$

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$$= \frac{1}{160}e^{2t}\sin(2t) + \frac{3}{160}e^{2t}\cos(2t)$$

Then the particular solution is

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(\frac{1}{240}e^{6t}\right)(e^{-3t}) + \left(-\frac{3}{160}e^{2t}\sin(2t) + \frac{1}{160}e^{2t}\cos(2t)\right)(e^t\cos(2t))$$

$$+\left(\frac{1}{160}e^{2t}\sin(2t) + \frac{3}{160}e^{2t}\cos(2t)\right)(e^t\sin(2t))$$

$$y_p(t) = \frac{1}{240}e^{3t} - \frac{3}{160}e^{3t}\sin(2t)\cos(2t) + \frac{1}{160}e^{3t}\cos^2(2t)$$

$$+\frac{1}{160}e^{3t}\sin^2(2t) + \frac{3}{160}e^{3t}\sin(2t)\cos(2t)$$

$$y_p(t) = \frac{1}{240}e^{3t} + \frac{1}{160}e^{3t}(\sin^2(2t) + \cos^2(2t))$$

$$y_p(t) = \frac{1}{240}e^{3t} + \frac{1}{160}e^{3t}$$

$$y_p(t) = \frac{1}{96}e^{3t}$$

And the general solution is the sum of the complementary solution and the particular solution,

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{-3t} + c_2e^t\cos(2t) + c_3e^t\sin(2t) + \frac{1}{96}e^{3t}$$

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